

DTSA 5746

Cryptographic Hash and Integrity Protection

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	Machine Learning Engineer (36.0 / 100)	27.0 / 100
SPECIALIZATION Applied Cryptography		

27

CURRENT AVG SCORE
across 6 job roles

59

POTENTIAL SCORE
with recommended additions

+42

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

This course describes the cryptographic hash functions and explains their security properties enabling their use for integrity protection. The course will also describe hash chain and Merkle tree. We will study message authenticity and the message authentication code (MAC) based on symmetric keys. Then, we will discuss digital signatures based on asymmetric cryptography, providing security objectives of non-repudiation which were unavailable in the symmetric-cryptography-based MAC.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	36.0	57.2	+21.2	MEDIUM
AI Engineer	35.8	66.3	+30.5	HIGH
Applied Scientist	25.0	64.3	+39.3	HIGH
ML Engineer	24.4	53.6	+29.1	MEDIUM
MLOps Engineer	20.4	48.1	+27.7	MEDIUM
Data Scientist	20.3	62.4	+42.1	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design

4 of 6 roles

systems

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1

HIGH

Add System Design coverage — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

GAP KEYWORDS BY ROLE

Machine Learning Engineer	36.0 → 57.2 (+21.2 pts)
data learning tools design science models model applications	
AI Engineer	35.8 → 66.3 (+30.5 pts)
data models learning build software model systems problems	
Applied Scientist	25.0 → 64.3 (+39.3 pts)
computational interdisciplinary problems models drug data learning systems	
ML Engineer	24.4 → 53.6 (+29.1 pts)
media data learning machine users systems models model	
MLOps Engineer	20.4 → 48.1 (+27.7 pts)
data model systems learning machine media tools models	
Data Scientist	20.3 → 62.4 (+42.1 pts)
definition data product different social build problems approaches	

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.